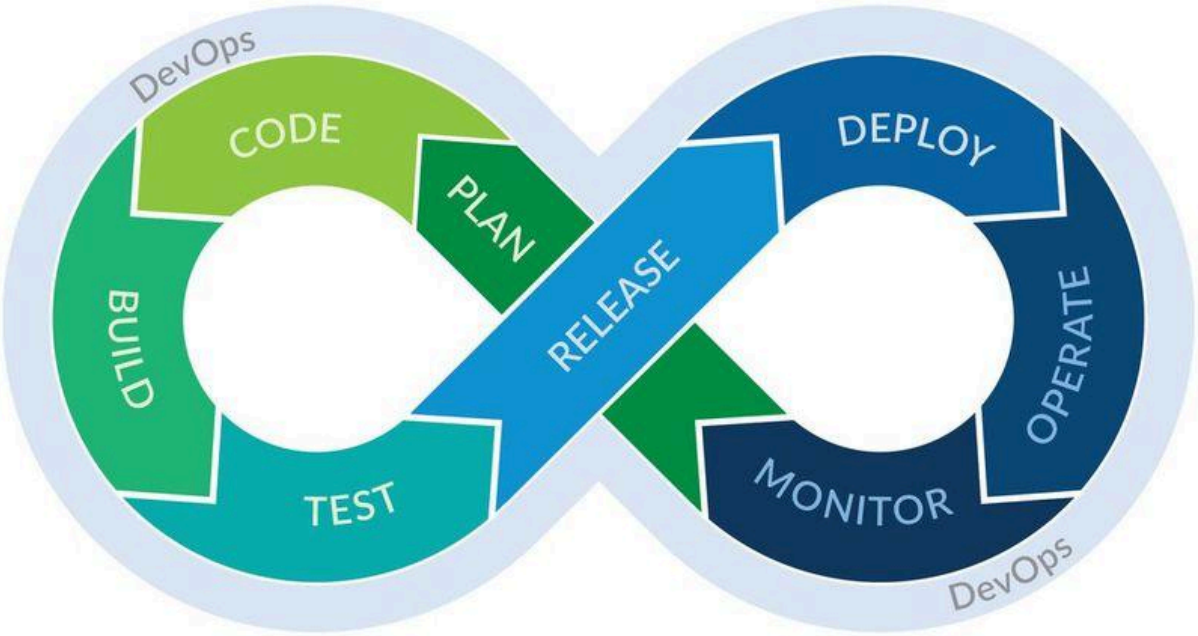


CI/CD and DevOps What is DevOps?

DevOps = Development + Operations

DevOps is a culture, process, and set of tools that help developers and operations teams work together to build, test, deploy, and maintain software faster and more reliably.

Before DevOps

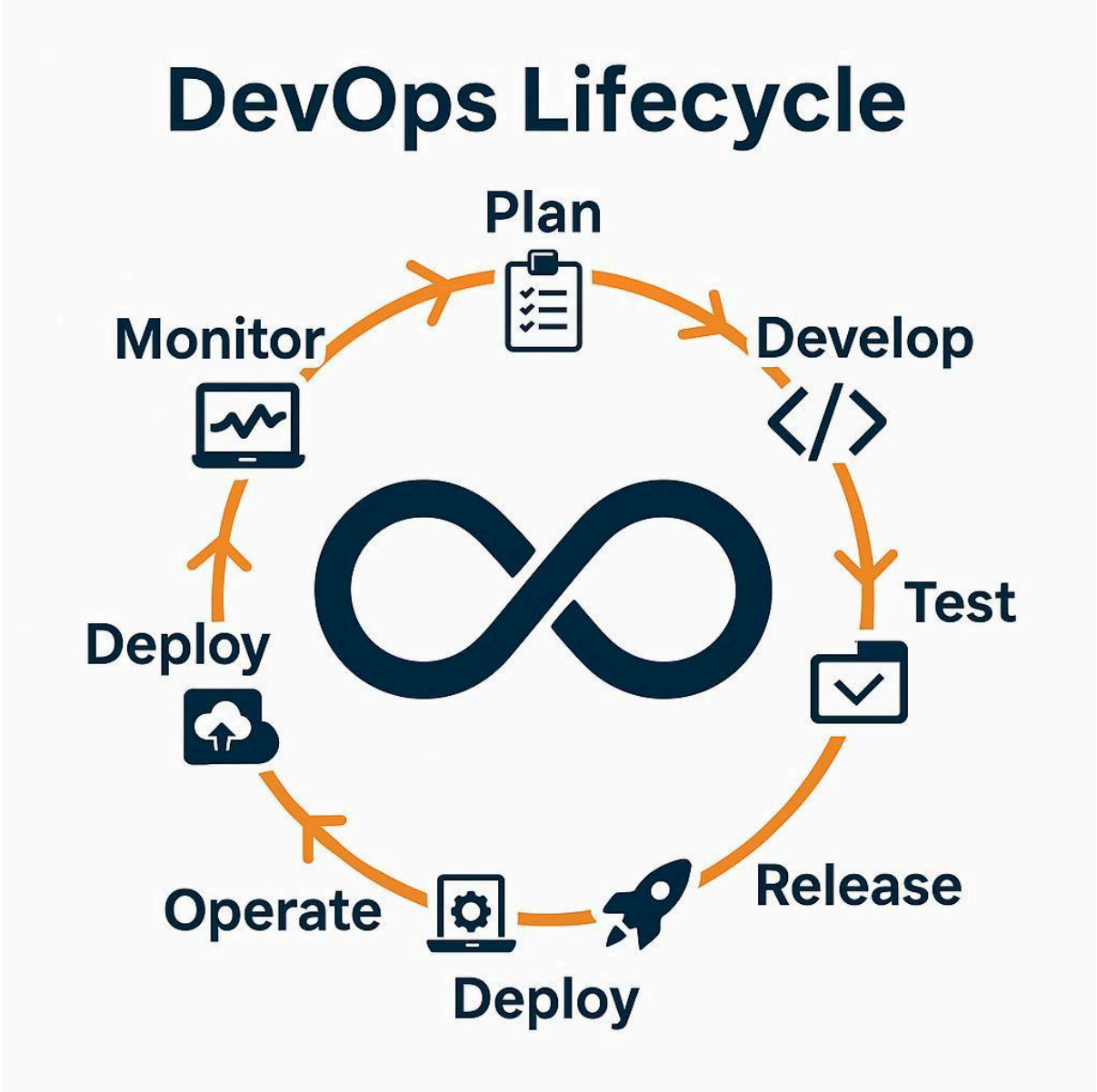
- Developers write code.
- Operations team deploys the application.
- Communication gaps occur.
- Deployment takes weeks or months.
- Many production failures.

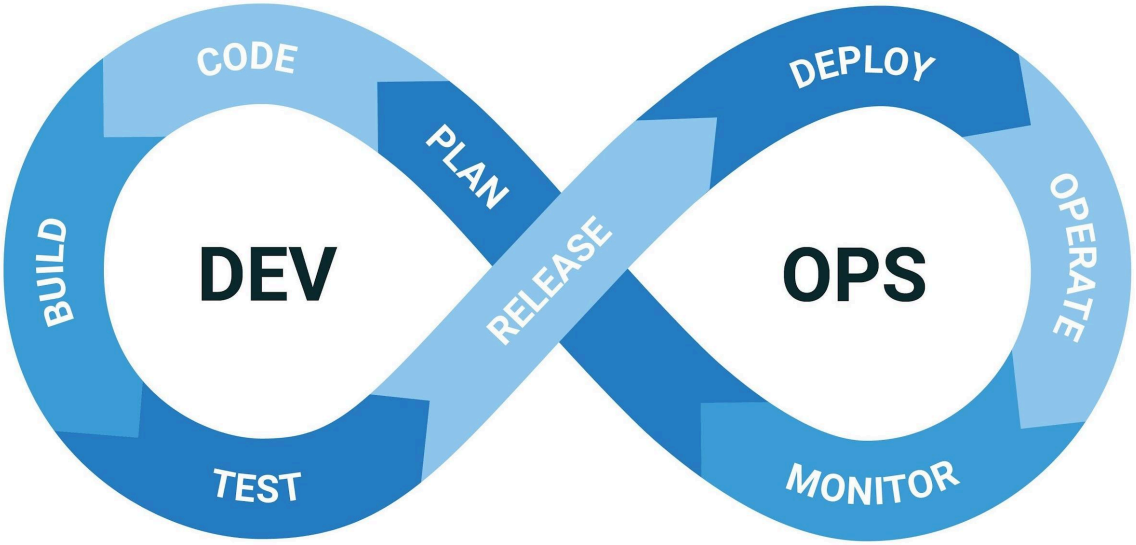
After DevOps

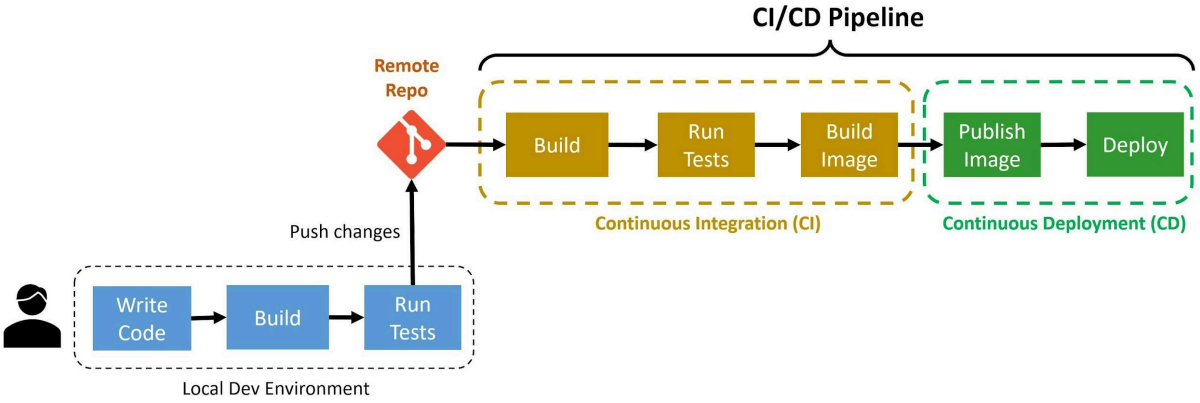
- Developers and Operations work together.
- Automated testing and deployment.

- Faster releases.
- Better quality software.
- Continuous monitoring.

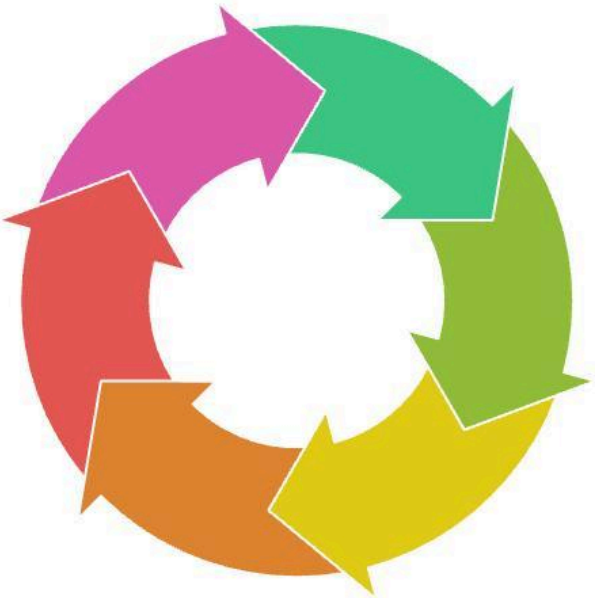
DevOps Workflow




CODE
PLAN
BUILD
DEV
TEST
RELEASE
DEPLOY
OPERATE
OPS
MONITOR


CI/CD Pipeline
Remote Repo
Build
Run Tests
Build Image
Publish Image
Deploy
Continuous Integration (CI)
Continuous Deployment (CD)
Push changes
Write Code
Build
Run Tests
Local Dev Environment

DevOps Deployment Cycle



Code Commit

Developers commit new code changes.

Build & Test

Code is built and tested automatically.

Package Application

Application is packaged for deployment.

Deploy to Staging

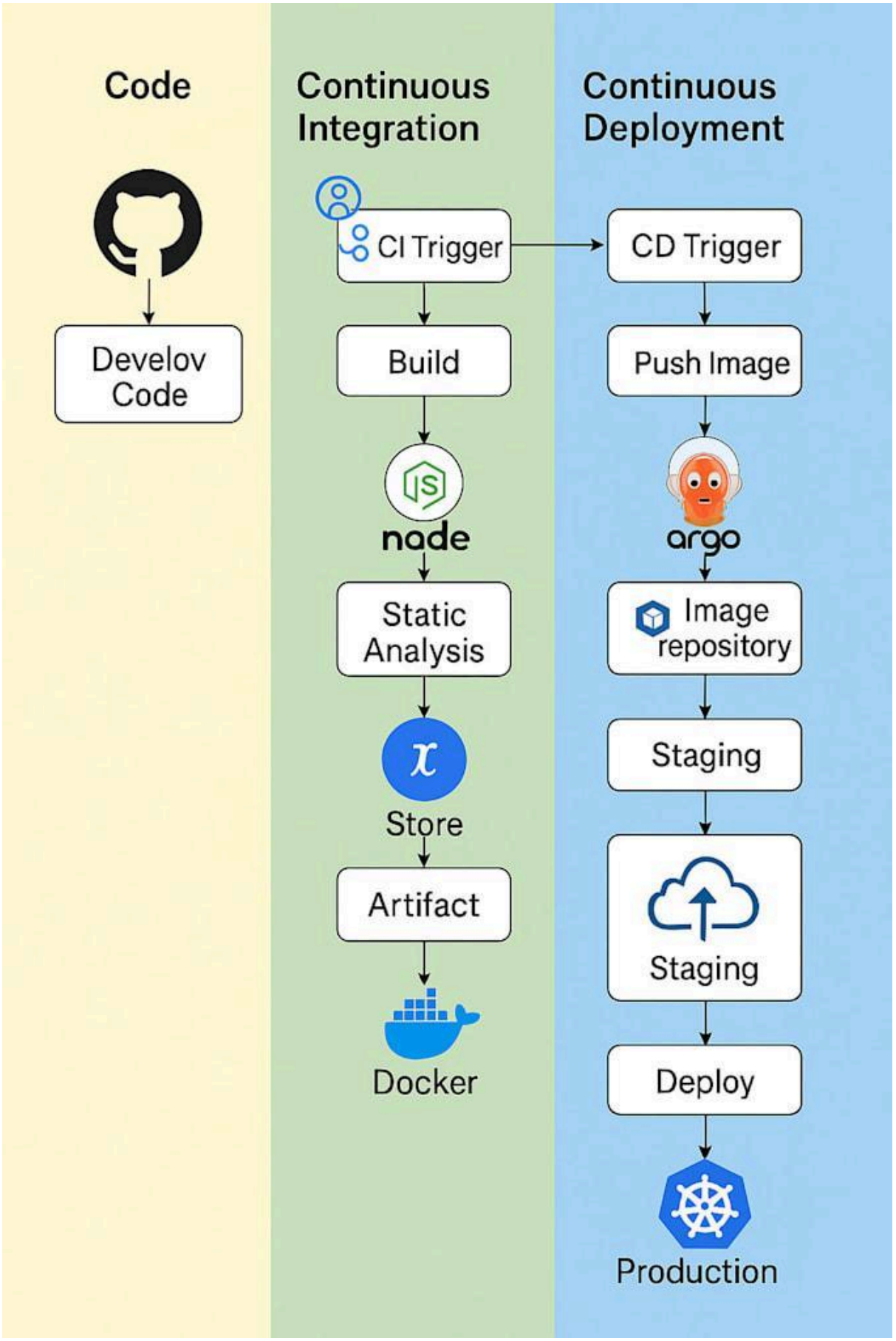
Application is deployed to staging environment.

Validate Release

Release is validated for production.

Deploy to Production

Application is deployed to production environment.


Code
Continuous Integration
Continuous Deployment
Develov Code
CI Trigger
Build
node
Static Analysis
Store
Artifact
Docker
CD Trigger
Push Image
argo
Image repository
Staging
Staging
Deploy
Production

DevOps Lifecycle

Plan
↓
Develop
↓
Build
↓
Test
↓
Release
↓
Deploy
↓
Operate
↓
Monitor
↓
Feedback
↓
Plan Again

This cycle keeps repeating continuously.

What is CI/CD?

CI/CD is the backbone of DevOps.

CI = Continuous Integration

Developers continuously integrate code into a shared repository.

CD = Continuous Delivery / Continuous Deployment

The integrated code is automatically tested and deployed.

Real-Time Example

Imagine you're building Swiggy.

Developer 1

Creates Login Module.

Developer 2

Creates Payment Module.

Developer 3

Creates Restaurant Module.

All developers push code to Git.

Instead of waiting 1 month to combine everything:

Developer
↓
Git Repository
↓
CI Pipeline
↓
Automatic Testing
↓
Build
↓
CD Pipeline
↓
Deployment

Every code change is automatically verified.

CI Pipeline (Continuous Integration)

What happens in CI?

Step 1: Developer Writes Code

```
def add(a,b):
    return a+b
```

Developer pushes code to Git.

Step 2: Git Detects Change

Tools:

- Git
- GitHub
- GitLab

Step 3: Build Process Starts

CI Server receives notification.

Popular Tools:

- Jenkins
- GitHub Actions
- GitLab CI/CD

Step 4: Run Automated Tests

Login Test
Payment Test
Security Test
Performance Test

If tests fail:

❌ Deployment stops.

If tests pass:

✅ Move to next stage.

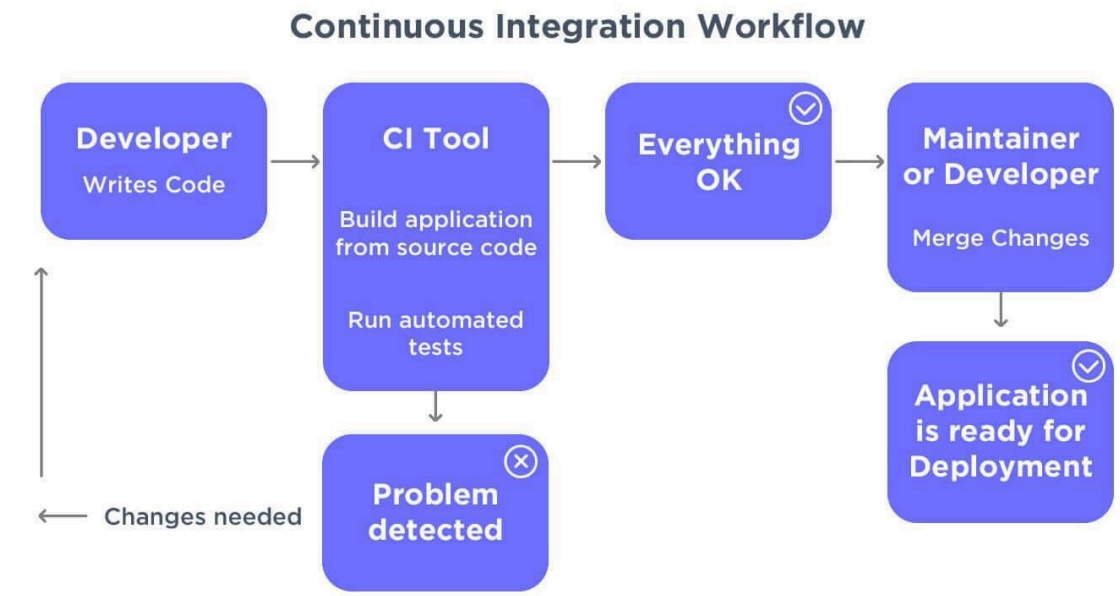
Step 5: Create Build Artifact

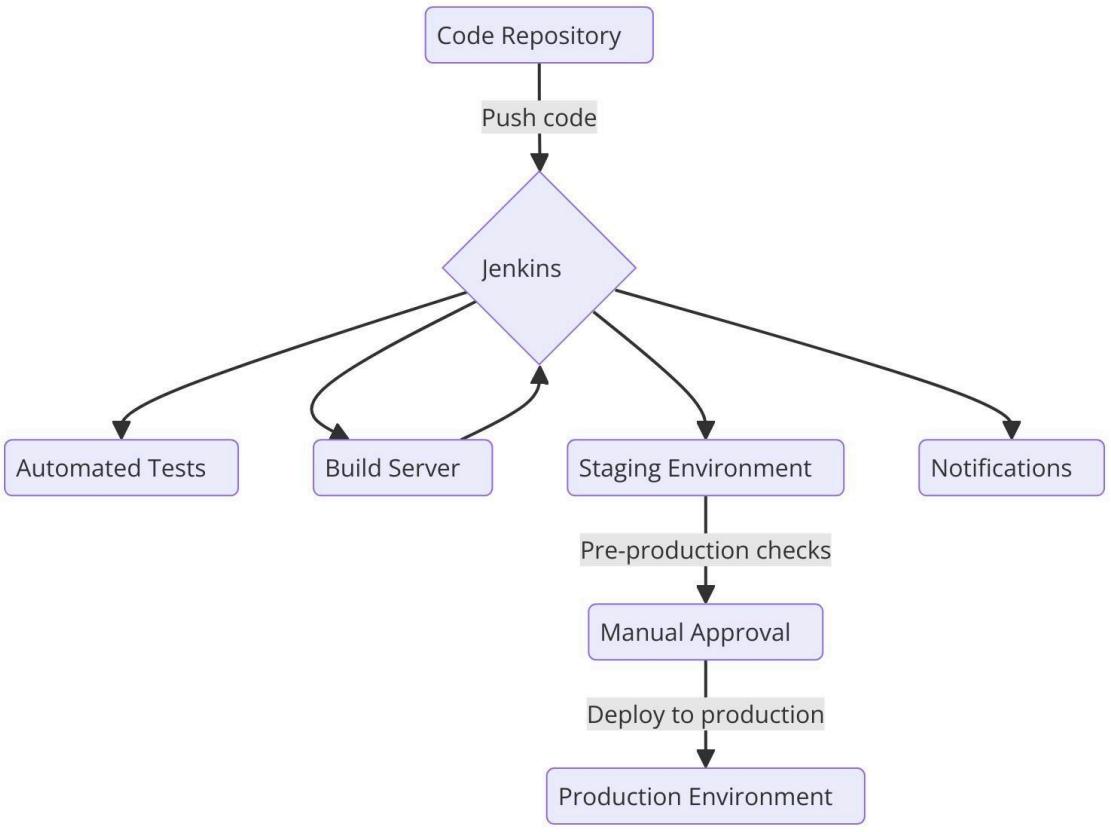
Examples:

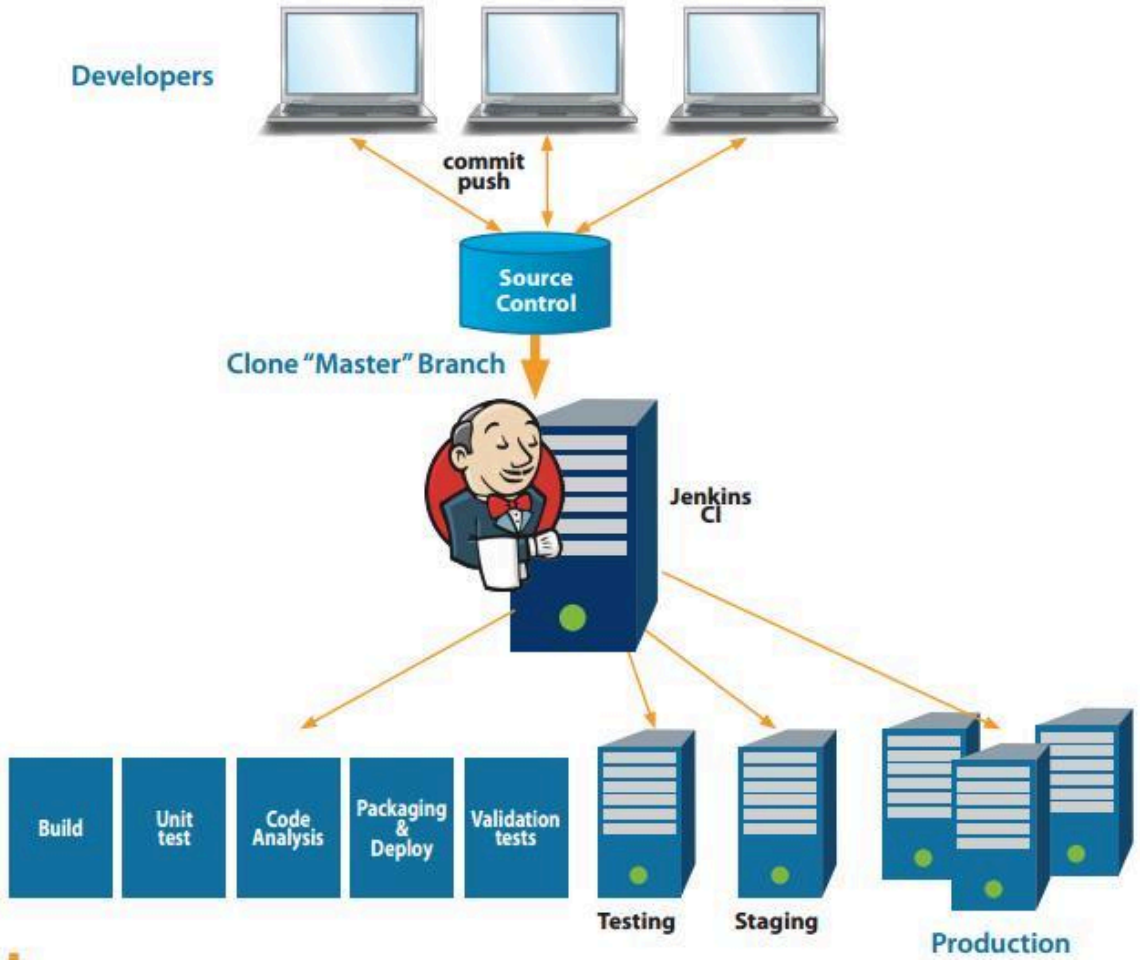
.jar
.war
.exe
Docker Image

Build is stored.

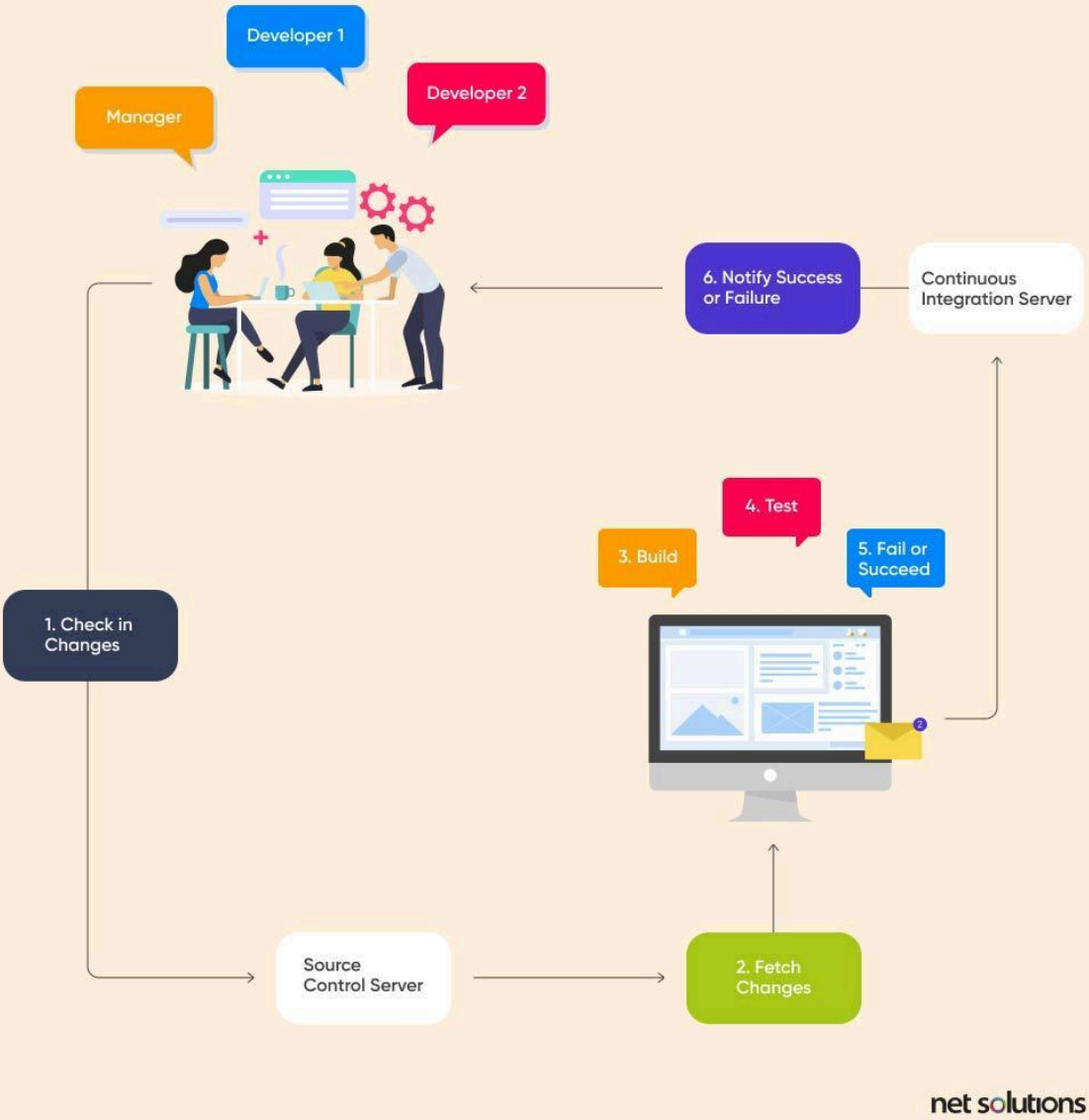
CI Workflow Image



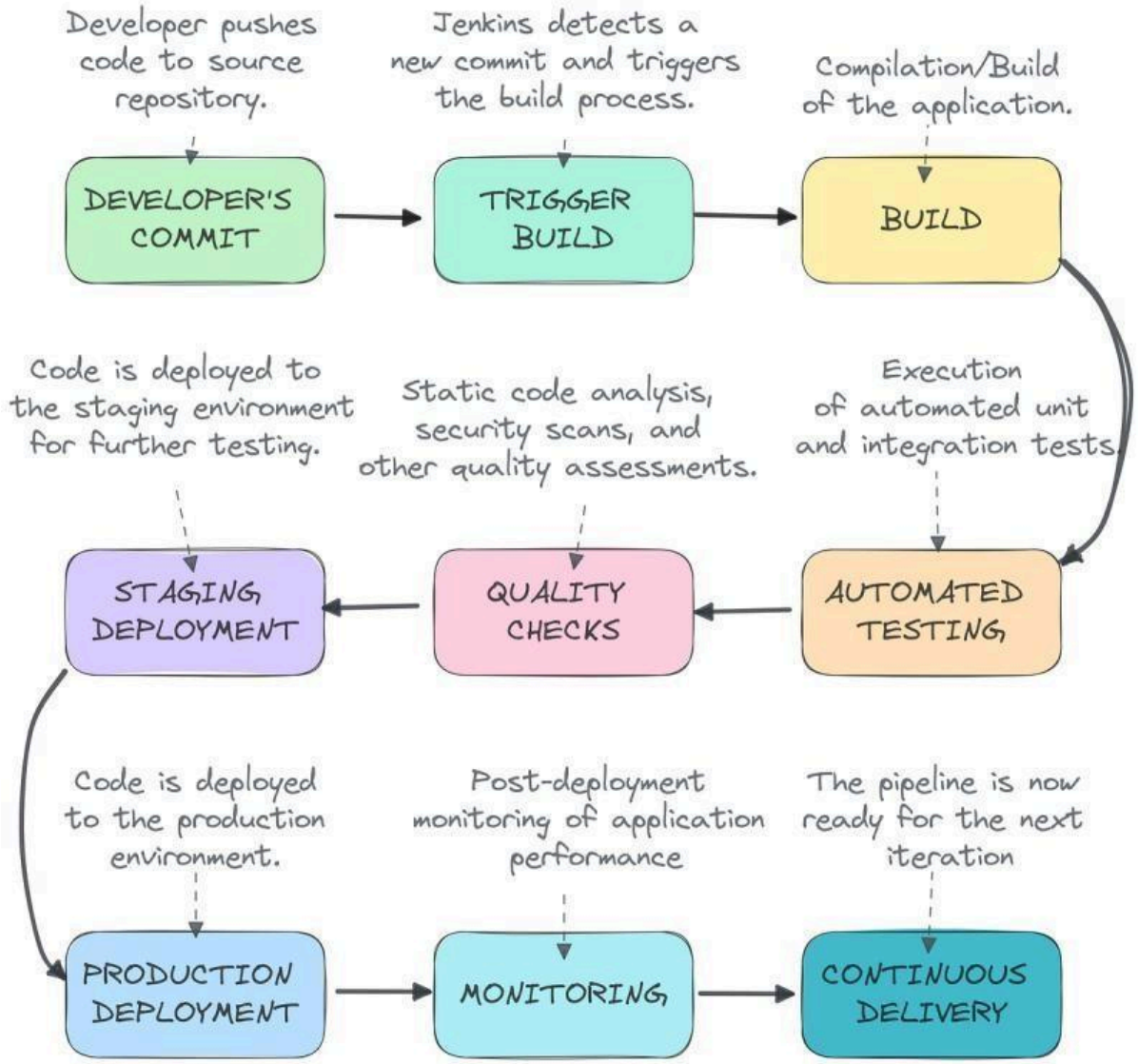



Developers
commit
push
Source
Control
Clone "Master" Branch
Jenkins
CI
Build
Unit
test
Code
Analysis
Packaging
&
Deploy
Validation
tests
Testing
Staging
Production

The Continuous Integration Process



net solutions



Developer
↓
Git Commit
↓
Build
↓
Unit Test
↓
Integration Test
↓
Artifact Creation

CD Pipeline (Continuous Delivery / Deployment)

Once CI succeeds, CD starts.

What Happens in CD?

Step 1

Take Build Artifact

Docker Image
or
JAR File

Step 2

Deploy to Testing Environment

Development
↓
Testing
↓
Staging
↓
Production

Step 3

Run Additional Tests

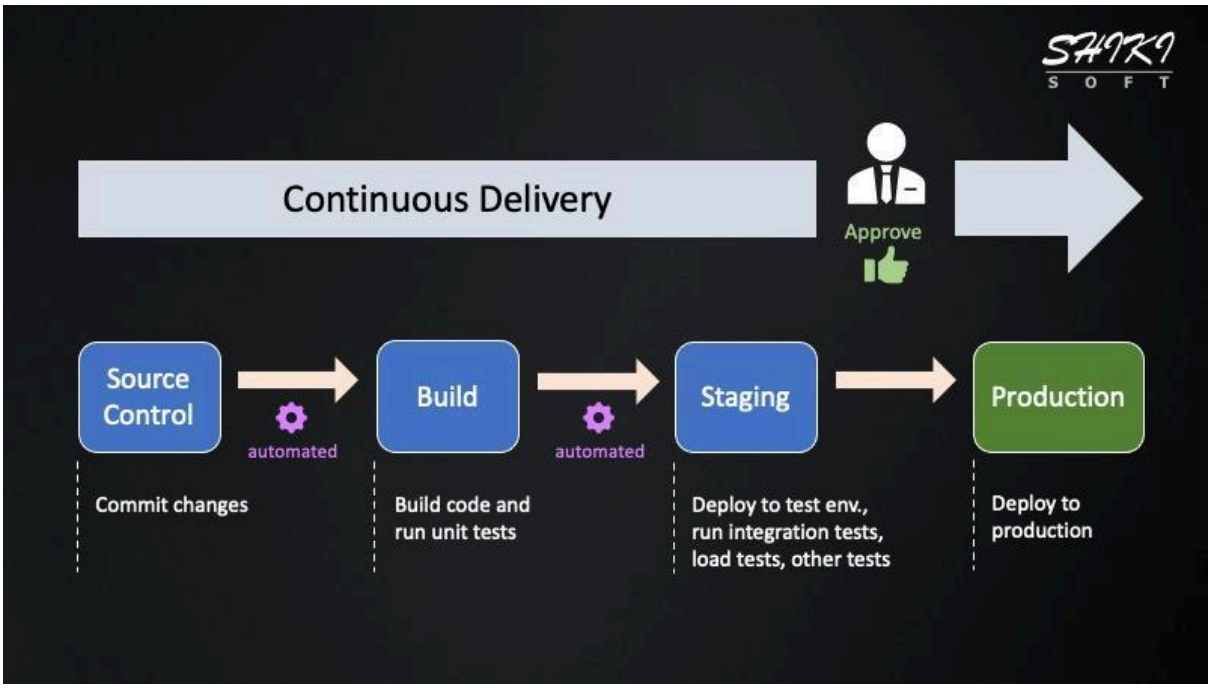
- Functional Testing
- Security Testing
- Load Testing

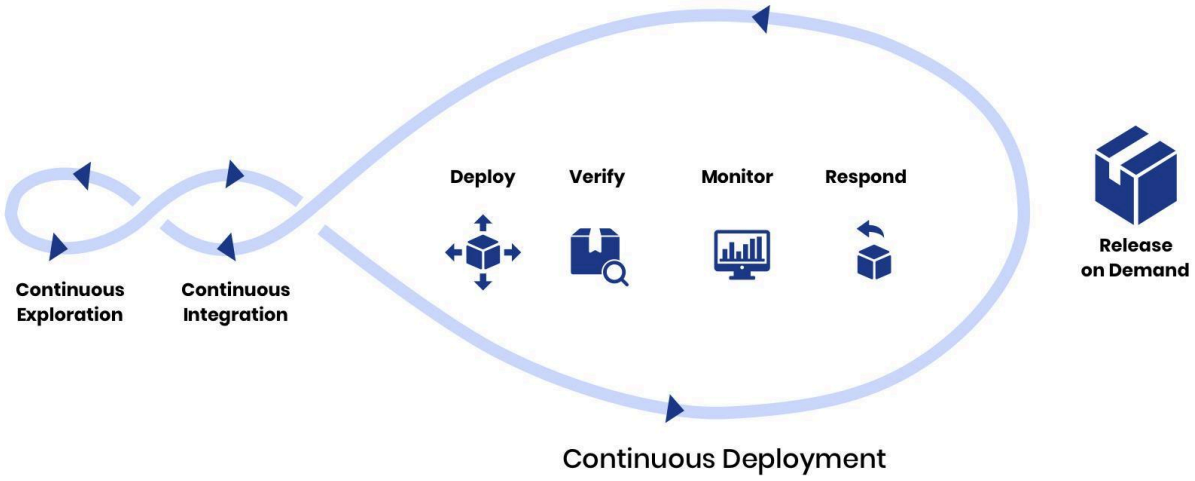
Step 4

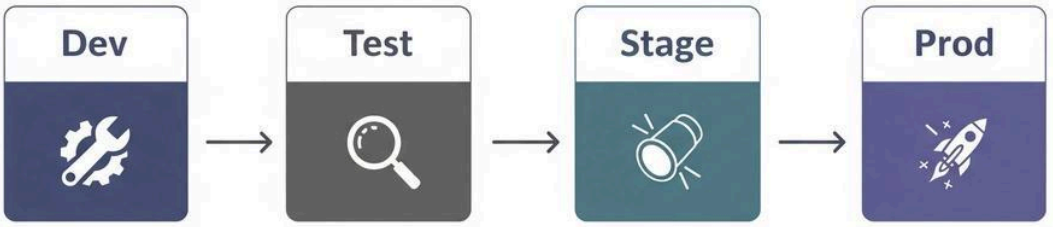
Deploy to Production

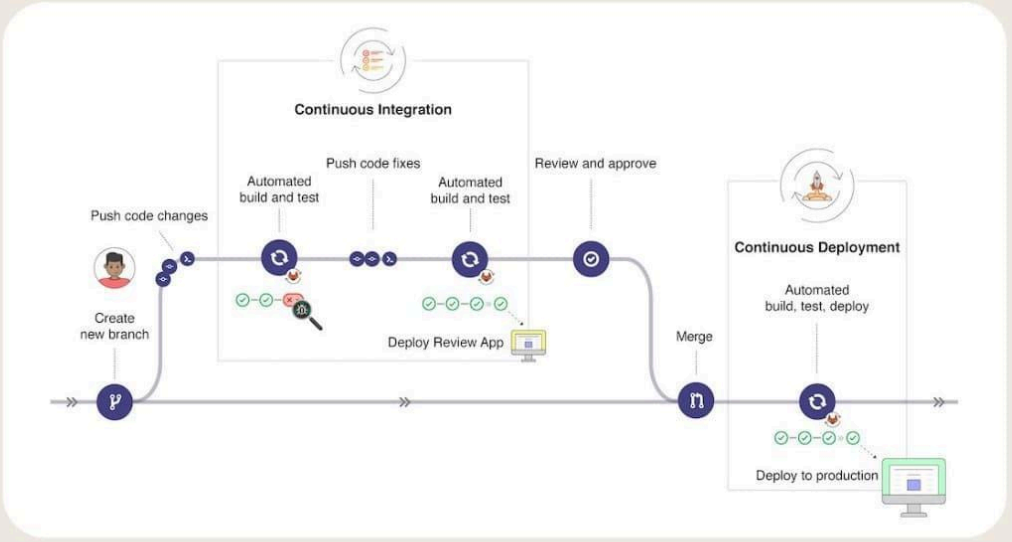
Users can now access the application.

CD Workflow Image

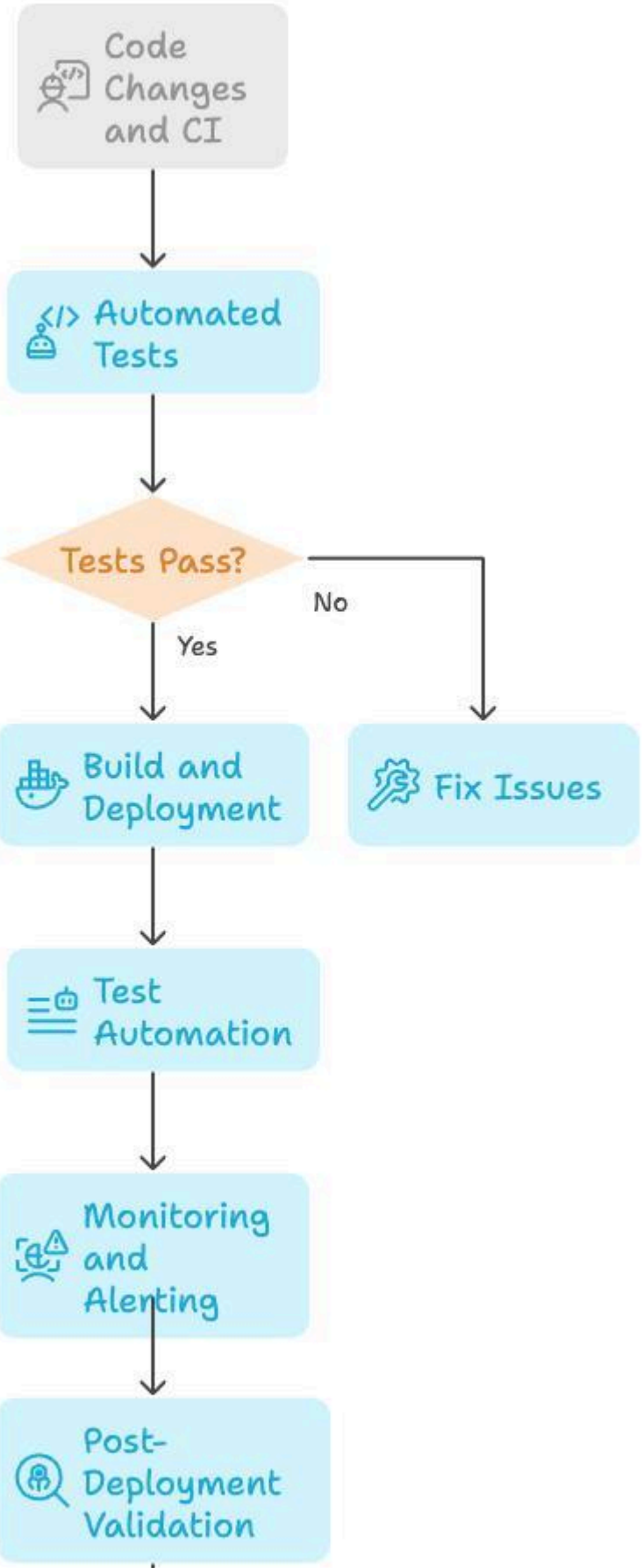





Dev
Test
Stage
Prod


Continuous Integration
Push code fixes
Review and approve
Automated build and test
Automated build and test
Push code changes
Continuous Deployment
Automated build, test, deploy
Create new branch
Deploy Review App
Merge
Deploy to production

Continuous Deployment Process



Artifact
↓
Testing Server
↓
Staging Server
↓
Production Server

How Data Moves Through CI/CD

Many students think CI/CD transfers business data.

Actually:

CI/CD primarily moves code.

Example:

Source Code
Configuration Files
Docker Images
Build Artifacts
Deployment Scripts

Flow:

Developer
↓
Git Repository
↓
Build Server
↓
Artifact Repository
↓
Testing Server
↓
Production Server

What Does a DevOps Engineer Do?

Daily Activities

Morning

Check:

- Production Health
- Server Status
- Application Status

Tools:

- Grafana
- Prometheus

During Development

Create CI/CD Pipelines.

Example:

Push Code
↓
Run Tests
↓
Build
↓
Deploy

Infrastructure Management

Manage:

- Servers
- Cloud
- Databases
- Networking

Cloud Platforms:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud

Container Management

Use:

- Docker
- Kubernetes

Monitoring

Track:

CPU Usage
Memory Usage
Disk Usage
Network Traffic
Application Errors

Real-Time DevOps Engineer Workflow

Scenario: Swiggy New Feature Release

Developer

Creates coupon feature.

Code Written

↓

Push to GitHub

↓

CI Pipeline

Compile
Test
Build

↓

DevOps Engineer

Create Docker Image
Store Image
Deploy to Kubernetes

↓

CD Pipeline

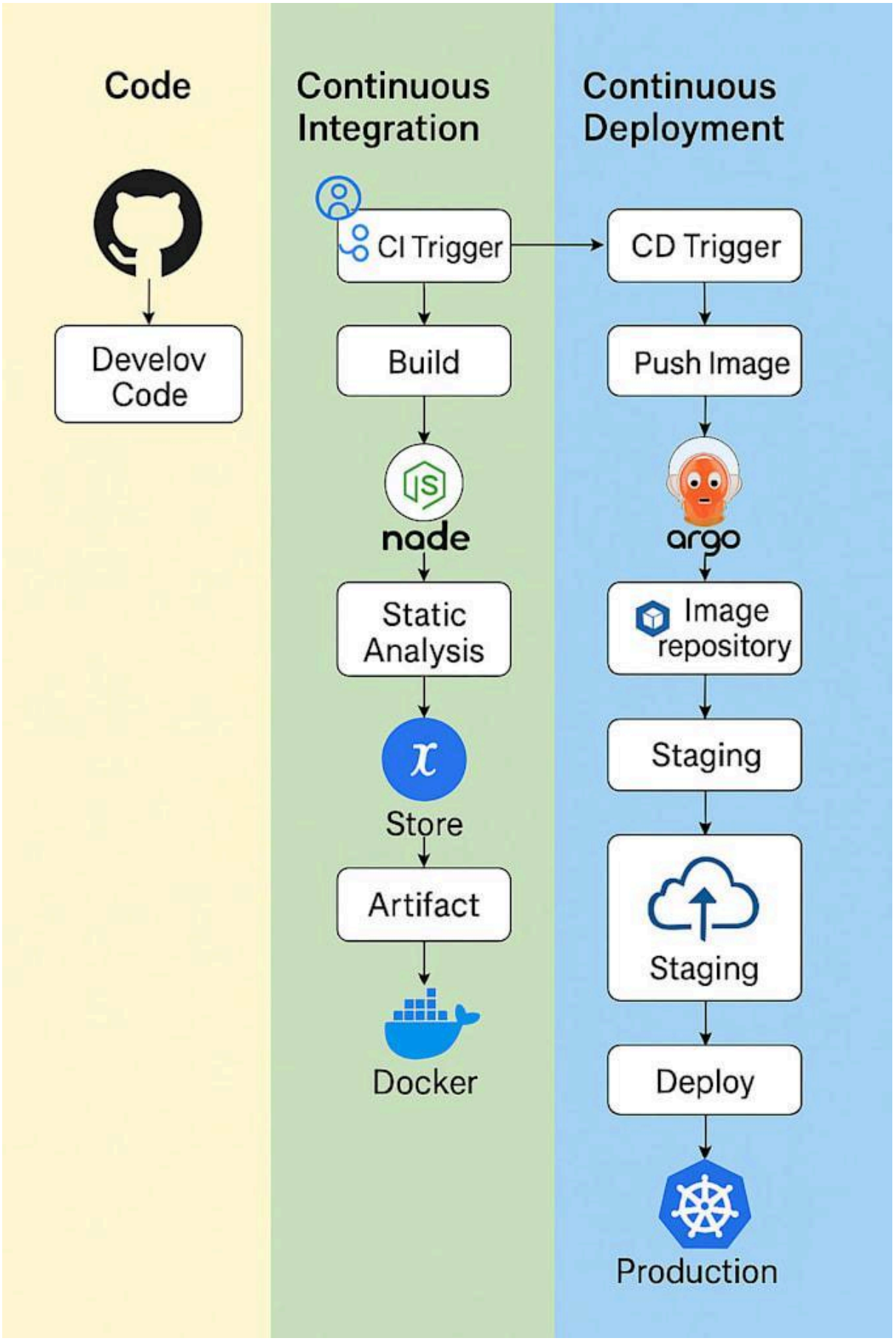
Testing
Staging
Production

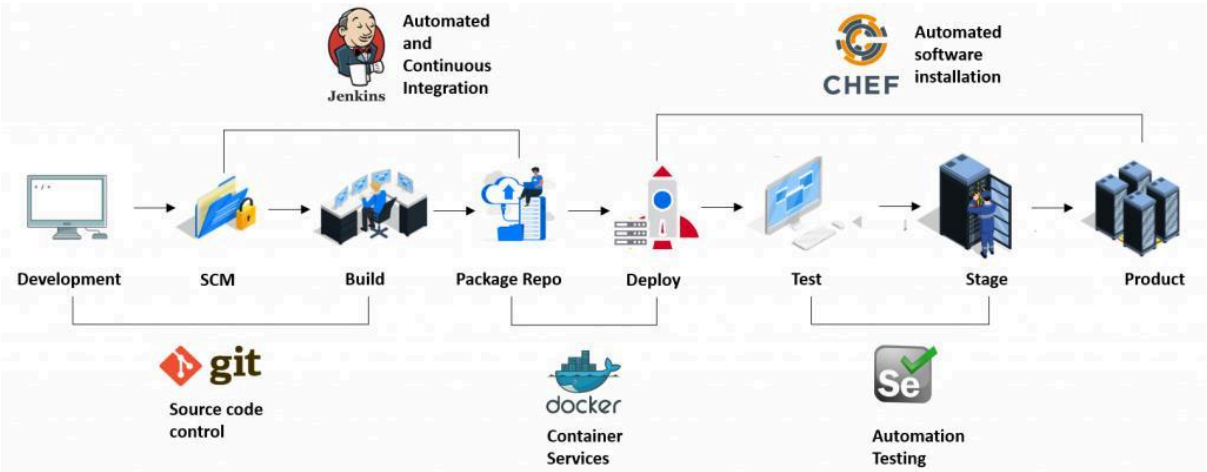
↓

Users

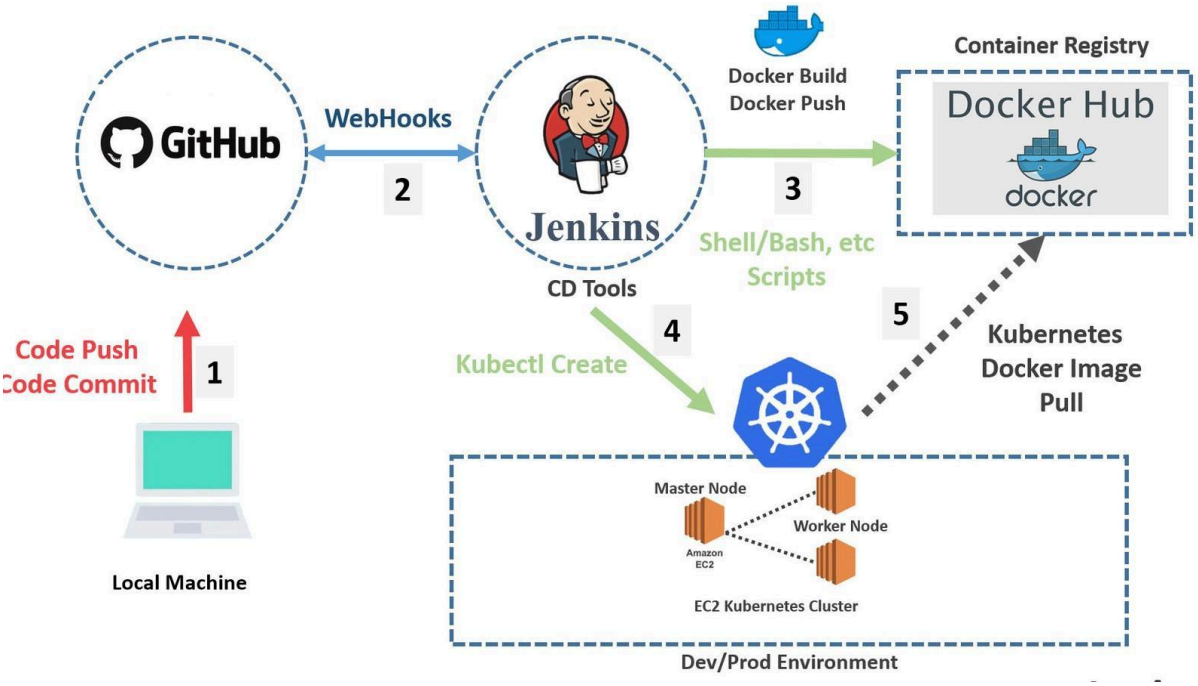
Use the new feature.

Complete CI/CD Architecture


Code
Continuous Integration
Continuous Deployment
Develov Code
CI Trigger
Build
node
Static Analysis
Store
Artifact
Docker
CD Trigger
Push Image
argo
Image repository
Staging
Staging
Deploy
Production

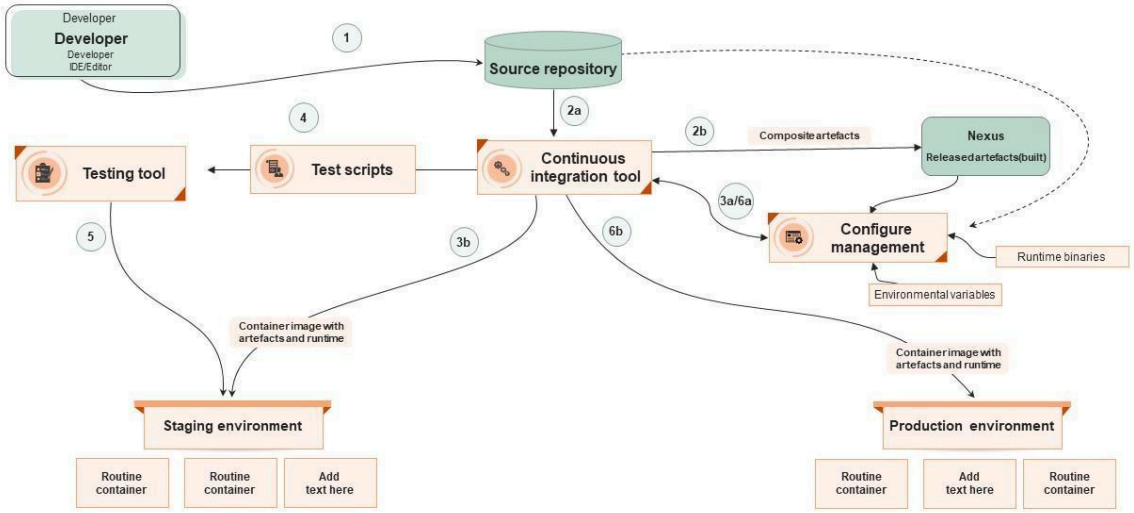


Typical Architecture of DevOps Life Cycle

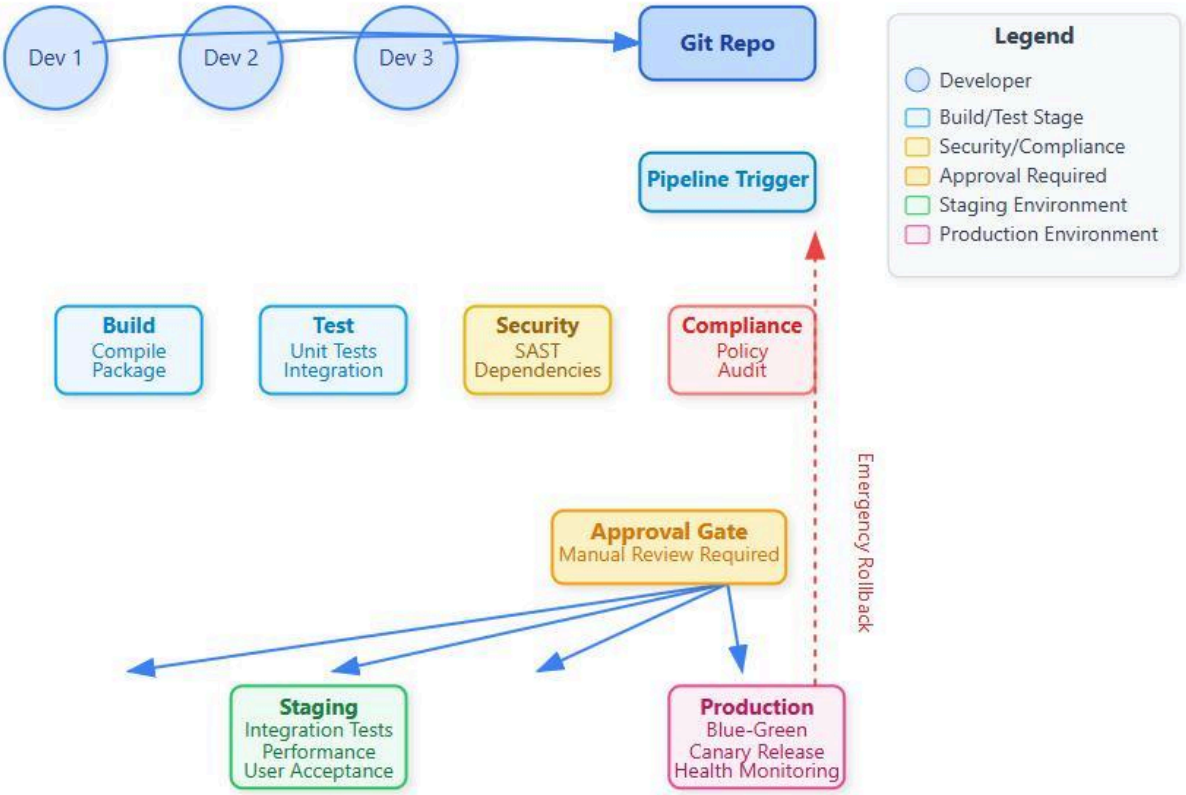


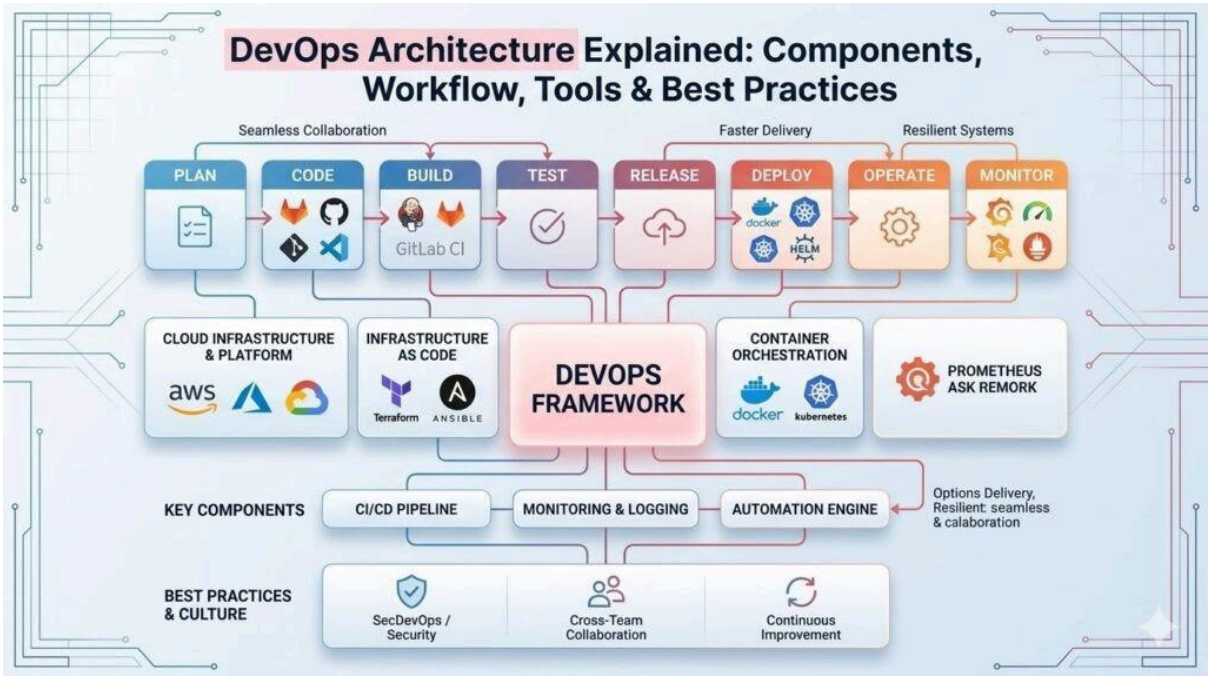
Enterprise CICD process implementation workflow

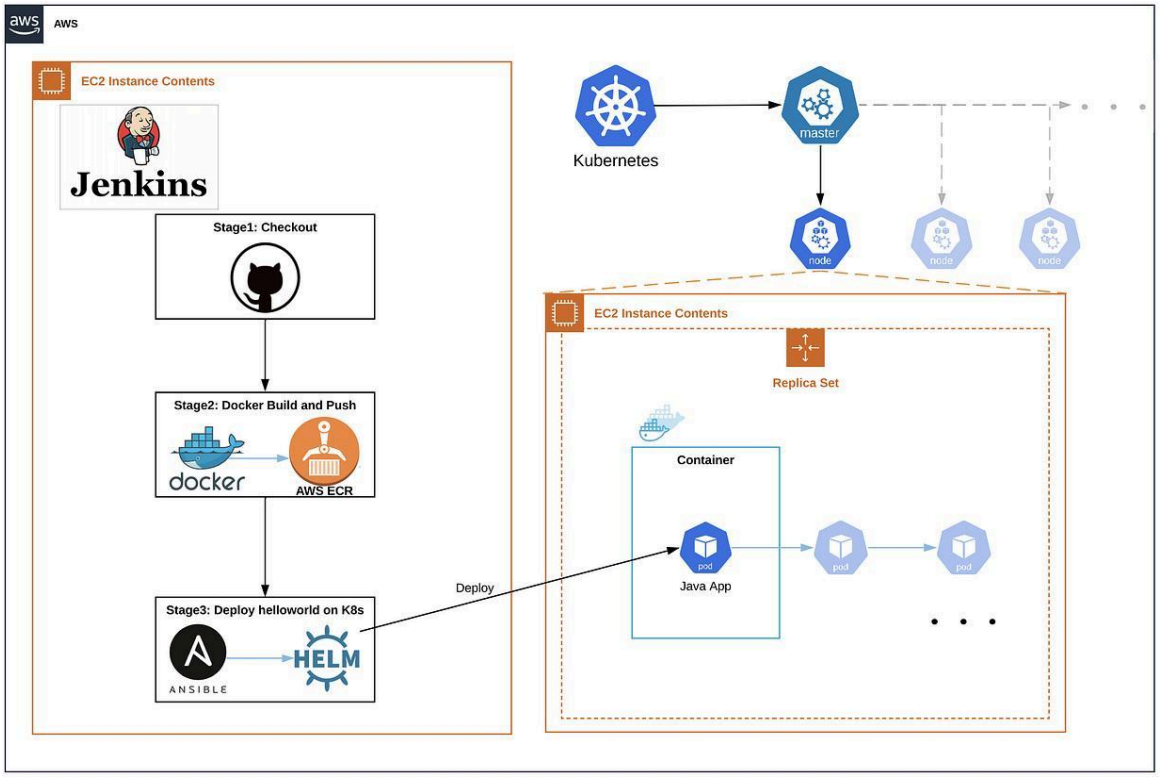
This slide represents the workflow of implementing CICD in an enterprise to help organization to scale in engineering team size, codebase size and infrastructure. It starts with developer, source repository and ends with container image with artifacts and runtime.



Enterprise CI/CD Pipeline Architecture







Developer

↓

GitHub

↓

Jenkins

↓

Build
↓
Test
↓
Docker
↓
Artifact Repository
↓
Kubernetes
↓
Production Server
↓
Users

Most Popular DevOps Tools

Purpose	Tool
Source Control	Git
Repository	GitHub
CI/CD	Jenkins
Container	Docker
Orchestration	Kubernetes
Monitoring	Grafana
Monitoring	Prometheus
Cloud	Amazon Web Services
Cloud	Microsoft Azure
